David Bardalez

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EDUCATION

Olin College of Engineering May 2028
B.S. in Electrical and Computer Engineering Needham, MA
Coursework: Linear Algebra, Multivariable Calculus, Physics, Signals and Systems, Microelectronic Circuits, Machine Learning, Sensors, and Instrumentation

Technical Skills

Proficiencies: PCB Design, Embedded Systems, Sensor Integration, Board Bring-Up and Debug, Soldering (SMD/TH), Sequencing Logic, Computer Vision
Languages: Python, C/C++, MATLAB, Java
Tools: KiCad, SolidWorks, LTspice, Git, QGroundControl, Oscilloscope, Logic Analyzer, DMM

EXPERIENCE

Olin Electric Motorsports (Formula SAE) Needham, MA
High Voltage Systems Sep 2025 - Present
Precharge/Discharge PCB

- Designed a PCB for the Formula SAE high-voltage battery system to enable safe startup and shutdown
- Developed two-stage relay sequencing and protection circuitry to limit inrush current and reduce estimated precharge time by over 2×
- Implemented control logic and timing features to ensure reliable switching and reduce false triggering
- Collaborated with mechanical and software teammates to integrate the board into the vehicle's accumulator and controls architecture

OCTAL Student Research Team Needham, MA
Firmware Subteam Member Feb 2026 - Present

- Brought up SPI communication between STM32-based hardware and the ADBMS battery-monitoring path, validating register readback during board bring-up
- Implemented per-cell voltage readout over SPI and multiplexed thermistor readout using I2C-controlled MAX14661 muxes and STM32 ADC inputs
- Validated thermistor voltage acquisition and supported CAN integration for the broader drone electrical system

PROJECTS

Masterpiece — Chessboard Hall-Effect Sensing System Oct 2025 - Dec 2025

- Built a full-size sensor-based chessboard with 120 Hall-effect sensors to detect piece positions in real time and stream board state to a host device
- Designed custom PCBs for Hall-effect sensor arrays and I2C expansion to support scalable board-state detection
- Wrote embedded C/C++ firmware to configure expanders, poll sensor inputs, and map data into a board-state matrix
- Assembled and debugged the system to achieve full-board readout, identifying board flex and magnet alignment as key integration challenges

Basketball Drone Vision Independent Project | Jun 2025 - Present

- Built a computer vision pipeline using YOLOv8 to detect basketballs and hoops from camera input
- Collected and labeled 2,000+ images, then evaluated model performance to improve shot-detection reliability over time
- Configured Pixhawk flight hardware in QGroundControl, including calibration, flight modes, and failsafes, for future system integration and testing

PyTanks (Python / Pygame Tank Game) Collaborative Software Project | Mar 2025 - May 2025

- Built a Wii Tanks-inspired game in Python using an MVC architecture to separate model, controller, and display responsibilities across a modular codebase
- Implemented tank movement, mouse-based aiming, projectile spawning, bullet bounce/collision logic, and wall interactions
- Contributed unit tests and debugging for core gameplay behaviors including kills, deaths, enemy respawning, bombs, and score tracking